



ClimateWins Weather Predictions

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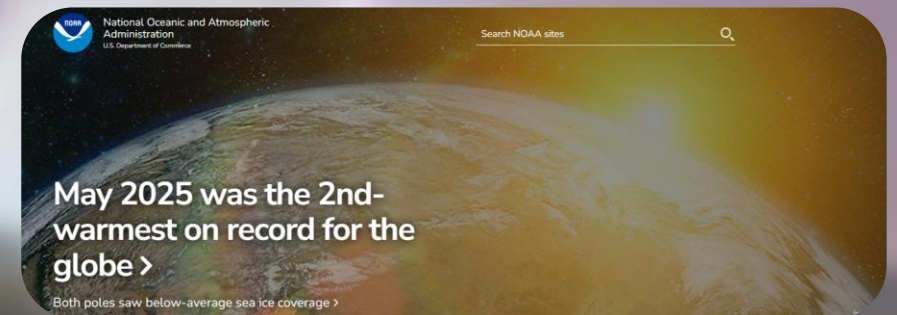
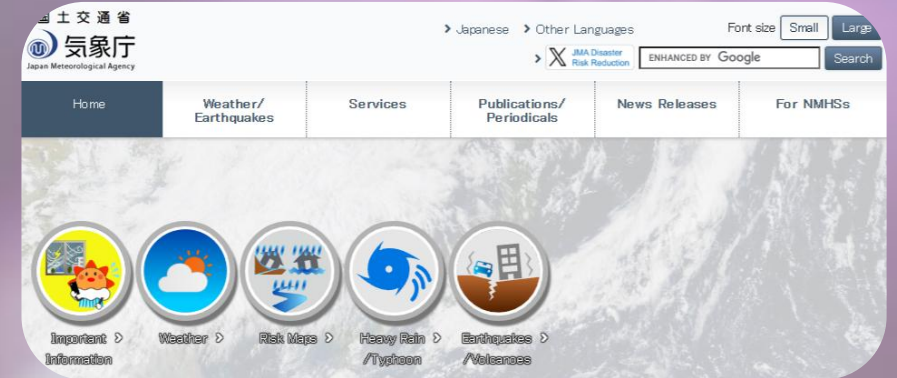
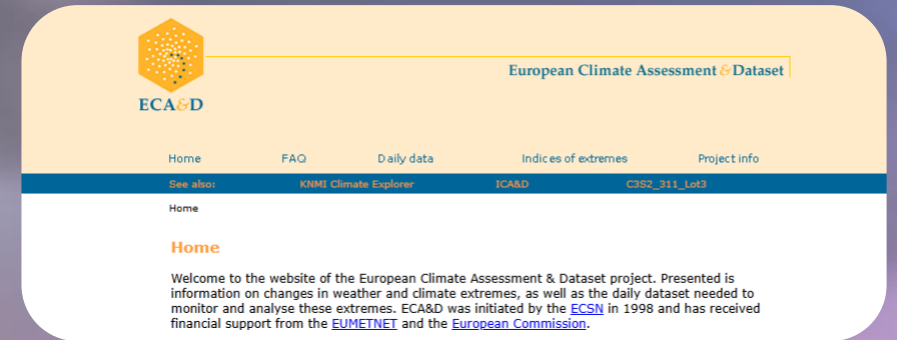
Objective & Data Sources

- Objective:

- Unveil capabilities of predicting climate change in Europe with Machine Learning.
- Working for a non-profit organization, I'm tasked with utilizing my skills as a data analyst, data scientist, researcher and a trainee. I'm trained on the job to learn and employ machine learning to help ClimateWins achieve its goal of understanding weather and its effects in Europe.

- Data Source:

- [European Climate Assessment & Data Set project](#)
- [The National Oceanic and Atmospheric Administration \(NOAA\)](#)
- [The Japan Meteorological Agency \(JMA\)](#)



Thought Experiments

1. **Can machine learning use historical weather data to predict weather trends for the following year(s)?**
2. **Can machine learning predict the amount of rain or snow that will accumulate based on cloud coverage, precipitation, and snow accumulation from prior seasons?**
3. **Can machine learning more accurately predict the improvement or decline of global warming?**

Algorithms Explored

Optimal Algorithms:

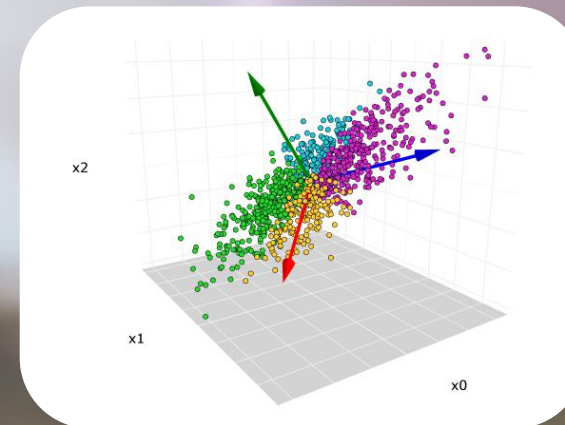
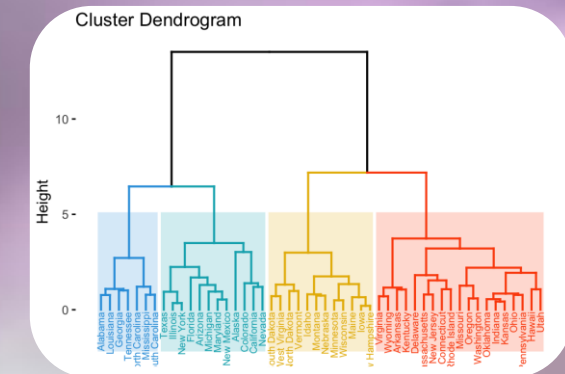
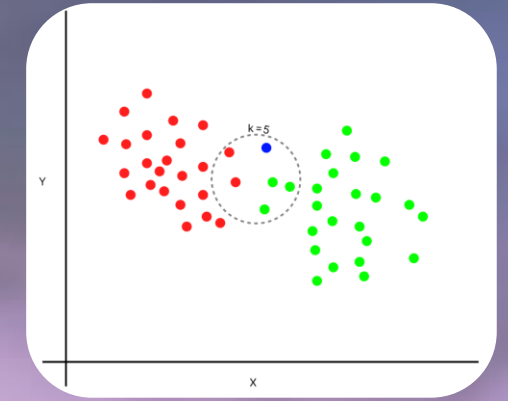
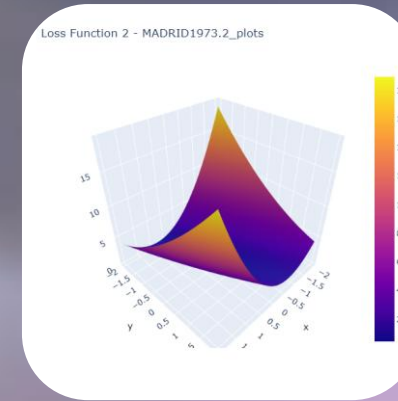
- Gradient Descent
 - A simple way to find a local minimum for linear and nonlinear cases.

Supervised Learning:

- K-Nearest Neighbor with Confusion Matrix
 - An instance-based algorithm used for classification and regression.
- Decision Tree
 - Classification-based algorithm where each decision narrows down a solution.
- Artificial Neural Networks (ANN)
 - Supervised learning that acts as unsupervised learning, this algorithm takes data and uses hidden layers to determine the output. This algorithm can determine patterns, classify data and predict outcomes.

Unsupervised Learning:

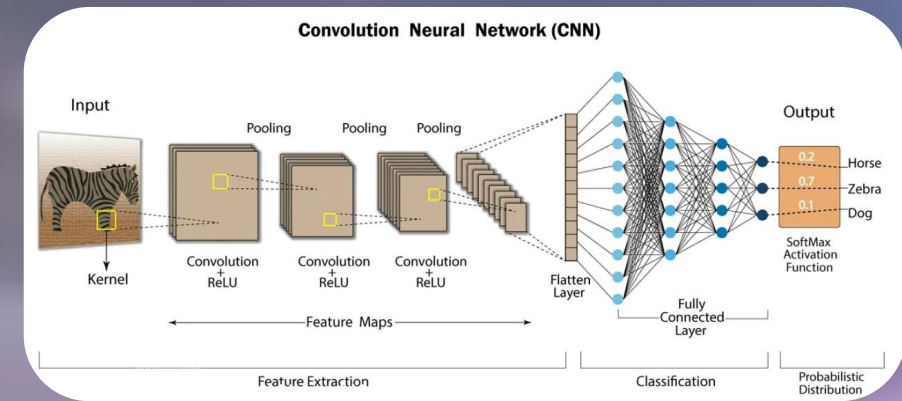
- Dendrograms
 - A type of agglomerative clustering used to visualize hierarchical relationships with data points or clusters.
- Principal Component Analysis (PCA)
 - Finds linear representation in data sets by looking at variances and turning the data into linear vectors.



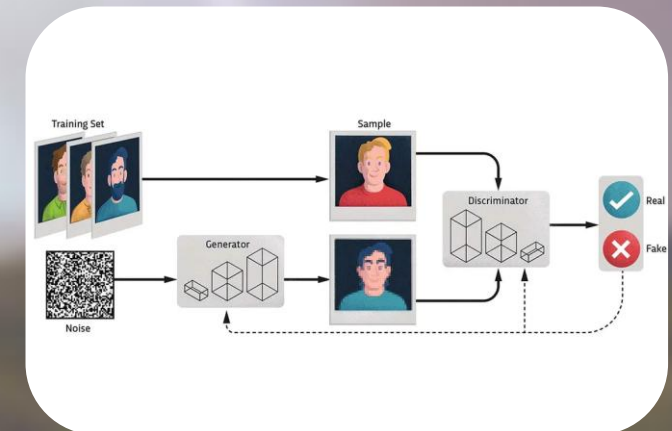
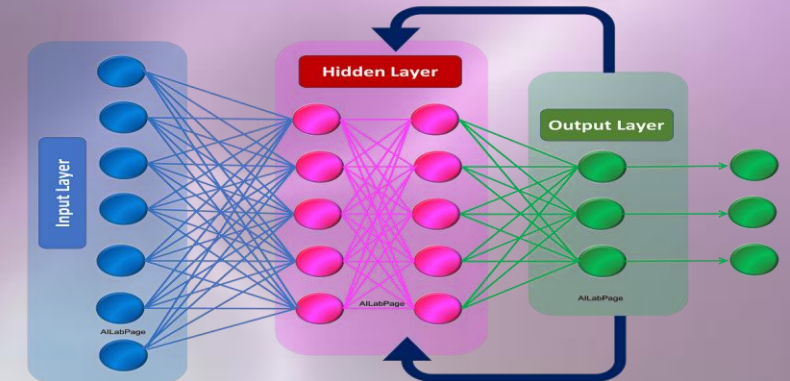
Algorithms Explored

Deep Learning:

- Convolution Neural Networks (CNN)
 - These models work well with images or numerical data, having hidden layers that can assess the data and become more complex as they communicate.
- Recurrent Neural Networks (RNN) with Long Short-Term Memory (LSTM)
 - A model that works well with text and speech it also contains a type of internal memory to feed its flow from input, hidden layer and output. LSTM are improved RNN that forgets unimportant data before continuing to process further.
- Random Forest
 - A collection of decision trees, this model trains many models simultaneously to solve a singular problem.
- Hyperparameters and Tuning models
 - Hyperparameters are the control of the overall machine learning models. Tuning models such as grid search, random search and Bayesian optimization allow for fine tuning of algorithms such as RNN and Random Forests.
- Generative Adversarial Network (GAN)
 - Using generators and discriminators, this visual machine learning algorithm tried to determine real from artificial data.



Recurrent Neural Networks

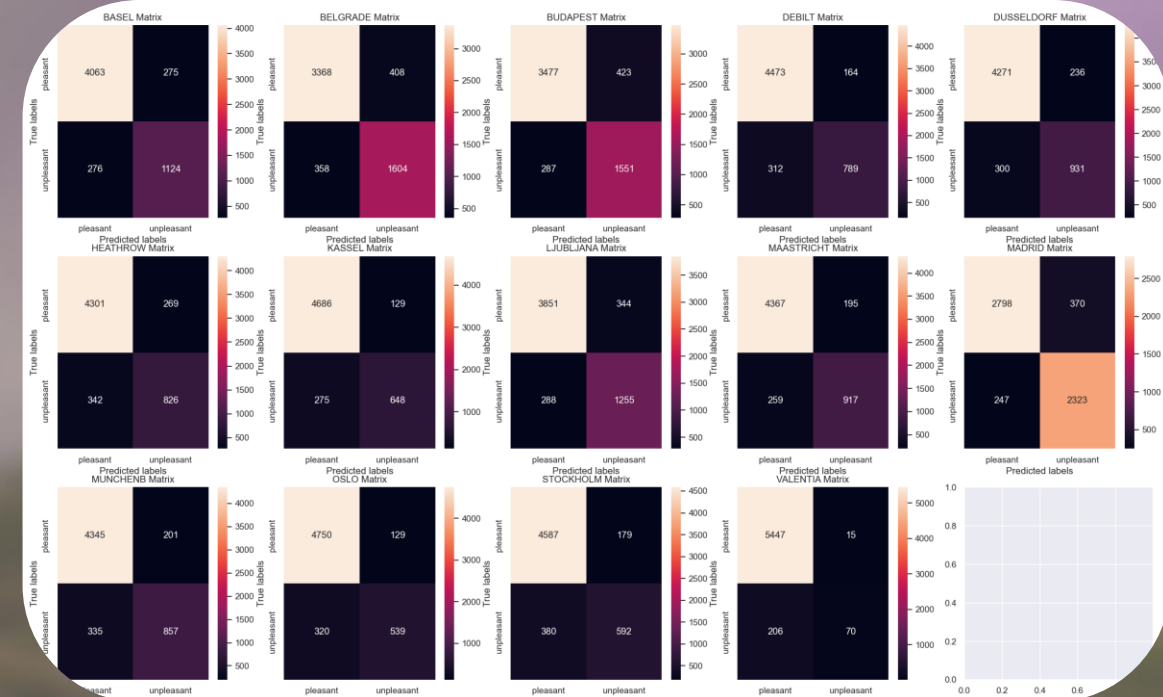
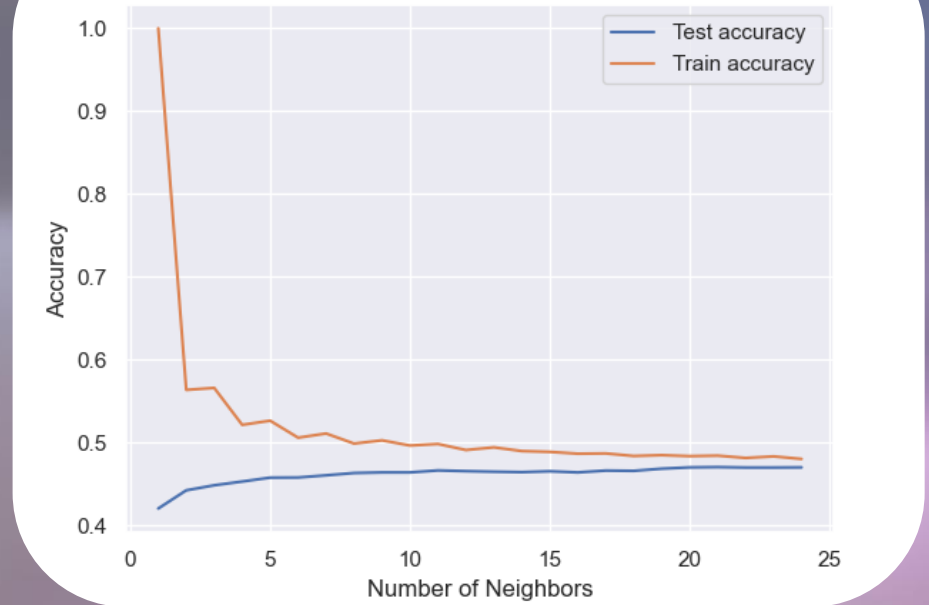


Machine Learning Most Applicable

- ClimateWins' goal is to predict the consequences of climate change in Europe.
- Options most likely to achieve this goal:
 - K-Nearest Neighbor (KNN)
 - Artificial Neural Networks (ANN)
 - Convolution Neural Networks (CNN)
 - Generative Adversarial Networks (GAN)

1. K- Nearest Neighbor with Confusion Matrix

- Adjusting the number of neighbors
- Pairing KNN with Confusion Matrices allow for better understanding of how pleasant and unpleasant weather is grouped.



Machine Learning Most Applicable

2. Artificial Neural Networks

- Adjust Hidden Layer Size
- Adjust Iterations
- Adjust Loss

ANN Attempt 2

```
[80]: # Create the ANN
# hidden_layer_sizes has up to three layers, each with a number of nodes. So (5, 5) is two hidden layers with 5 nodes each,
# and (100, 50, 25) is three hidden layers with 100, 50, and 25 nodes.
mlp2 = MLPClassifier(hidden_layer_sizes=(10, 5), max_iter=500, tol=0.0001)

# Fit the data to the model
mlp2.fit(X_train, y_train)

C:\Users\jmesi\anaconda3\Lib\site-packages\sklearn\neural_network\_multilayer_perceptron.py:690: ConvergenceWarning: Stochastic Optimizer: Maximum iterations (500) reached and the optimization hasn't converged yet.
warnings.warn(

[80]: ▼ MLPClassifier
MLPClassifier(hidden_layer_sizes=(10, 5), max_iter=500)

[83]: y_pred = mlp2.predict(X_train)
print(accuracy_score(y_pred, y_train))
y_pred_test = mlp2.predict(X_test)
print(accuracy_score(y_pred_test, y_test))

0.5223681152684174
0.5256186824677588
```

ANN Attempt 1

```
[68]: # Create the ANN
# hidden_layer_sizes has up to three layers, each with a number of nodes. So (5, 5) is two hidden layers with 5 nodes each,
# and (100, 50, 25) is three hidden layers with 100, 50, and 25 nodes.
mlp = MLPClassifier(hidden_layer_sizes=(5, 5), max_iter=500, tol=0.0001)

# Fit the data to the model
mlp.fit(X_train, y_train)

[68]: ▼ MLPClassifier
MLPClassifier(hidden_layer_sizes=(5, 5), max_iter=500)

[70]: y_pred = mlp.predict(X_train)
print(accuracy_score(y_pred, y_train))
y_pred_test = mlp.predict(X_test)
print(accuracy_score(y_pred_test, y_test))

0.4859981408319777
0.49477169745555943
```

ANN Attempt 3

```
] : # Create the ANN
# hidden_layer_sizes has up to three layers, each with a number of nodes. So (5, 5) is two hidden layers with 5 nodes each,
# and (100, 50, 25) is three hidden layers with 100, 50, and 25 nodes.
mlp3 = MLPClassifier(hidden_layer_sizes=(20, 10, 10), max_iter=1000, tol=0.0001)

# Fit the data to the model
mlp3.fit(X_train, y_train)

[90]: ▼ MLPClassifier
MLPClassifier(hidden_layer_sizes=(20, 10, 10), max_iter=1000)

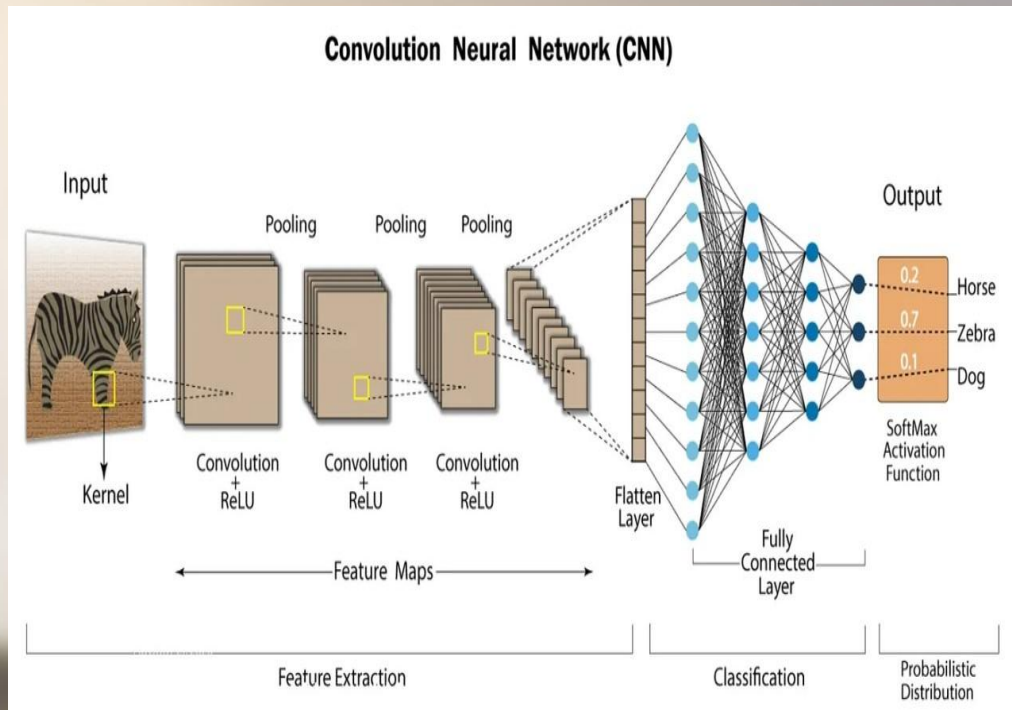
[92]: y_pred = mlp3.predict(X_train)
print(accuracy_score(y_pred, y_train))
y_pred_test = mlp3.predict(X_test)
print(accuracy_score(y_pred_test, y_test))

0.6706948640483383
0.6455210874869293
```


Machine Learning Most Applicable

3. Convolution Neural Networks

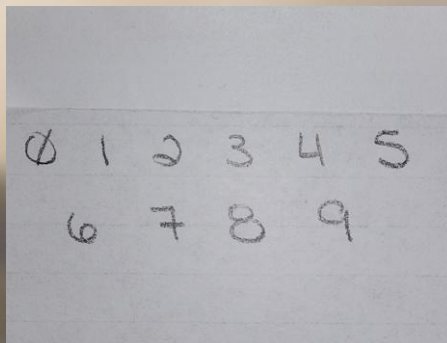
- This algorithm works well with images and numerical values.
- This algorithm paired with GAN or even amplified with hyperparameters could promise better predictability of unpleasant weather through imagery or text.



Machine Learning Most Applicable

4. Generative Adversarial Networks

- Generators create data based on real inputs
- Discriminator get samples of real and artificial data and try to determine which is which.



```
Predicting the Class

[540]: # Predict the class
       predict_value = model.predict(img0_new)
       digit = argmax(predict_value)
       print(digit)
       1/1 ————— 0s 25ms/step
       6

[542]: predict_value = model.predict(img1_new)
       digit = argmax(predict_value)
       print(digit)
       1/1 ————— 0s 20ms/step
       8

[544]: predict_value = model.predict(img2_new)
       digit = argmax(predict_value)
       print(digit)
       1/1 ————— 0s 19ms/step
       3

[546]: predict_value = model.predict(img3_new)
       digit = argmax(predict_value)
       print(digit)
       1/1 ————— 0s 20ms/step
       3

[548]: predict_value = model.predict(img4_new)
       digit = argmax(predict_value)
       print(digit)
       1/1 ————— 0s 19ms/step
       6

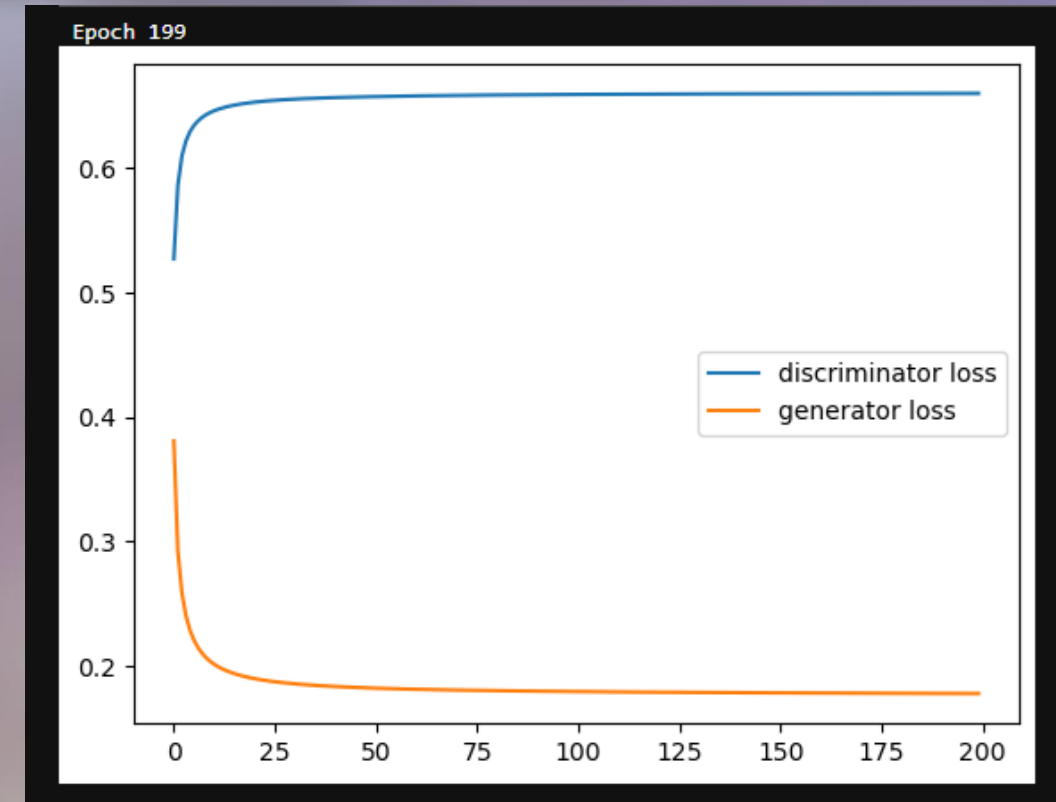
[550]: predict_value = model.predict(img5_new)
       digit = argmax(predict_value)
       print(digit)
       1/1 ————— 0s 18ms/step
       3

[552]: predict_value = model.predict(img6_new)
       digit = argmax(predict_value)
       print(digit)
       1/1 ————— 0s 19ms/step
       6

[554]: predict_value = model.predict(img7_new)
       digit = argmax(predict_value)
       print(digit)
       1/1 ————— 0s 19ms/step
       7

[556]: predict_value = model.predict(img8_new)
       digit = argmax(predict_value)
       print(digit)
       1/1 ————— 0s 20ms/step
       3

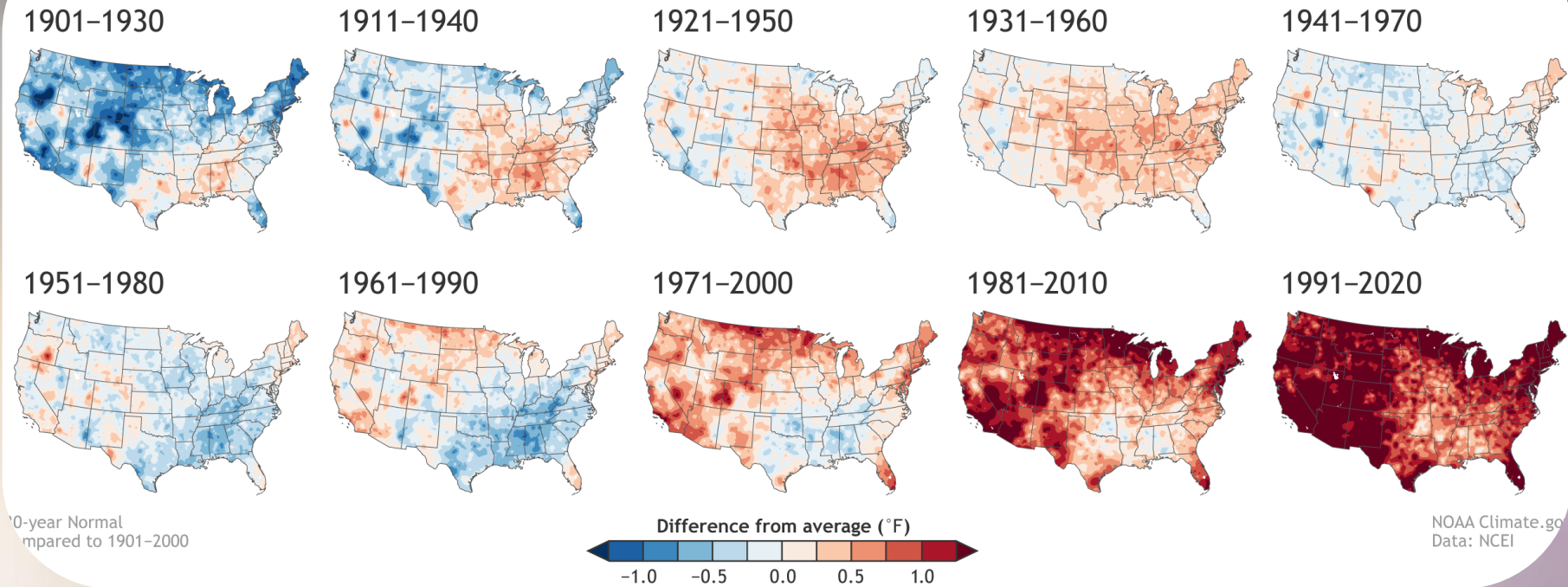
[558]: predict_value = model.predict(img9_new)
       digit = argmax(predict_value)
       print(digit)
       1/1 ————— 0s 19ms/step
       4
```



Historical Data

- Europe's Environmental Disasters:
 - **Flooding**
 - Cloud cover, humidity, pressure, precipitation
 - **Heat Wave**
 - Sunshine, Global radiation, temperature max
 - **Cold Spells**
 - Cloud cover, temperature min, snow depth
 - **Earthquakes**
 - No current data
 - **Landslides**
 - No current data

U.S. ANNUAL TEMPERATURE COMPARED TO 20th-CENTURY AVERAGE



Thought Experiment #1

1. Can machine learning use historical weather data to predict weather trends for the following year(s)?

- Machine Learning Method to Implement:
 - Artificial Neural Networks (ANN)
- Data:
 - Current data from 1960 - 2022

Thought Experiment #2



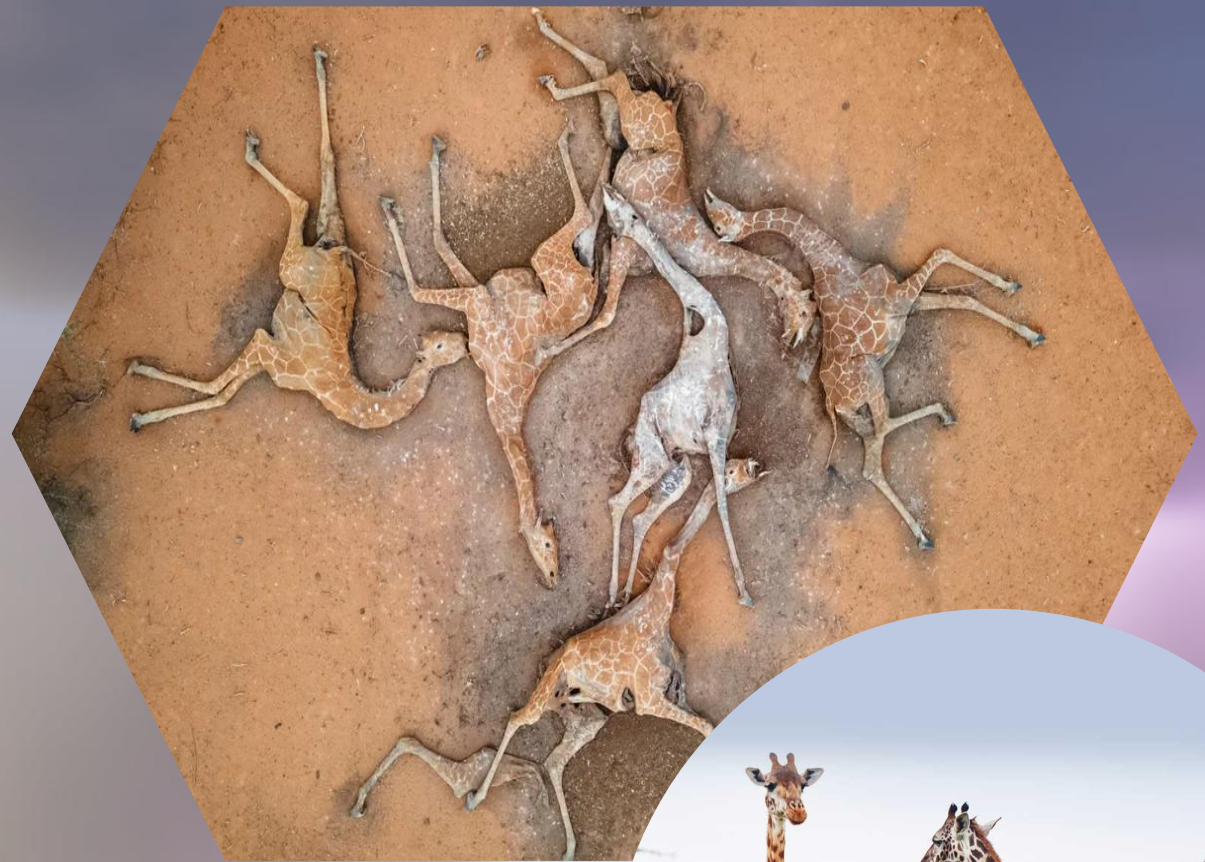
2. Can machine learning predict the amount of rain or snow that will accumulate based on cloud coverage, precipitation, and snow accumulation from prior seasons?

- Machine Learning Method to Implement:
 - Convolution Neural Network (CNN)
- Data:
 - Current data from 1960 – 2022
 - Specifically looking at cloud coverage, precipitation and snow accumulation.
 - Additionally looking at data containing freezing temperatures, winter storms and longer winters.

Thought Experiment #3

3. Can machine learning more accurately predict the improvement or decline of global warming?

- Machine Learning Method to Implement:
 - Gradient Descent
- Data:
 - Global weather data and event data.



Summary



- **Thought Experiment #1:**

- Can machine learning use historical weather data to predict weather trends for the following year(s)?
 - It's most likely to align with ClimateWins' goals.
 - Most easily executed.
 - Data is already present.
 - Next steps.

Thank You!

Thank you for joining me on this machine learning journey.

I'm happy to take questions from the audience.



Check out my work for this project on Github.



Check out my LinkedIn



Explore my website outlining my projects.



Jordan Novelli